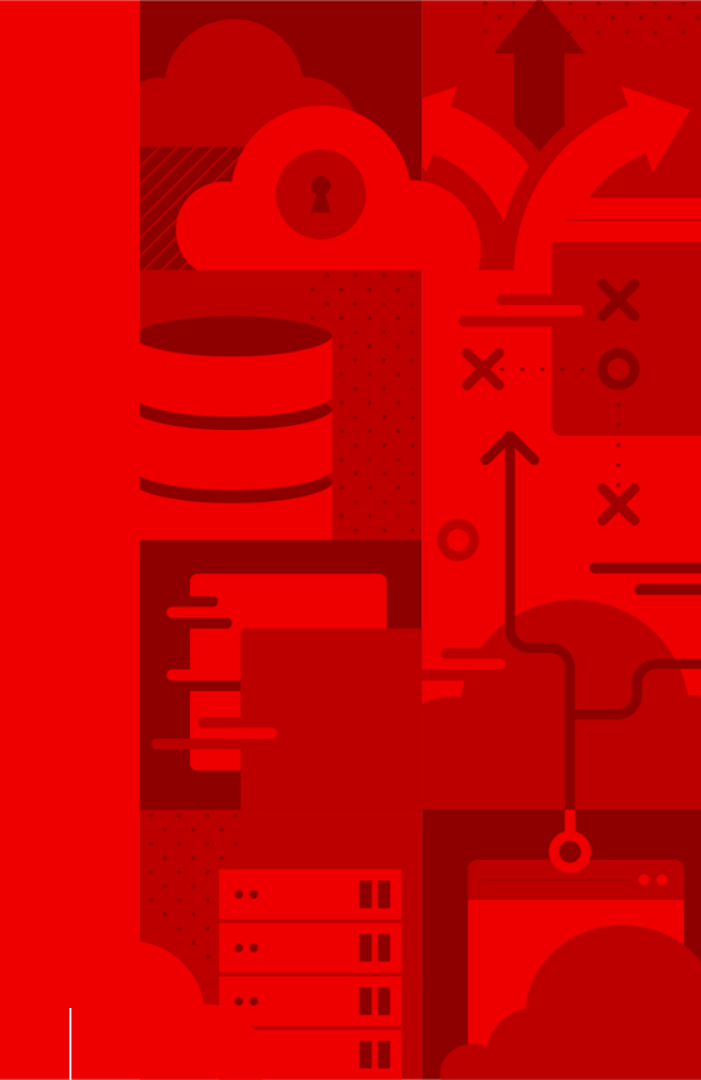


A vertical red graphic on the left side of the slide. It contains various white icons: a cloud with a keyhole, a database cylinder, a server rack, a web browser window, and several arrows pointing in different directions. There are also some 'X' and 'O' symbols and a circuit-like line.

Designing an Omnichannel Customer Experience

Red Hat Portfolio Architecture v3.2



Eric D. Schabell

Portfolio Architect Technical Director

May 2022



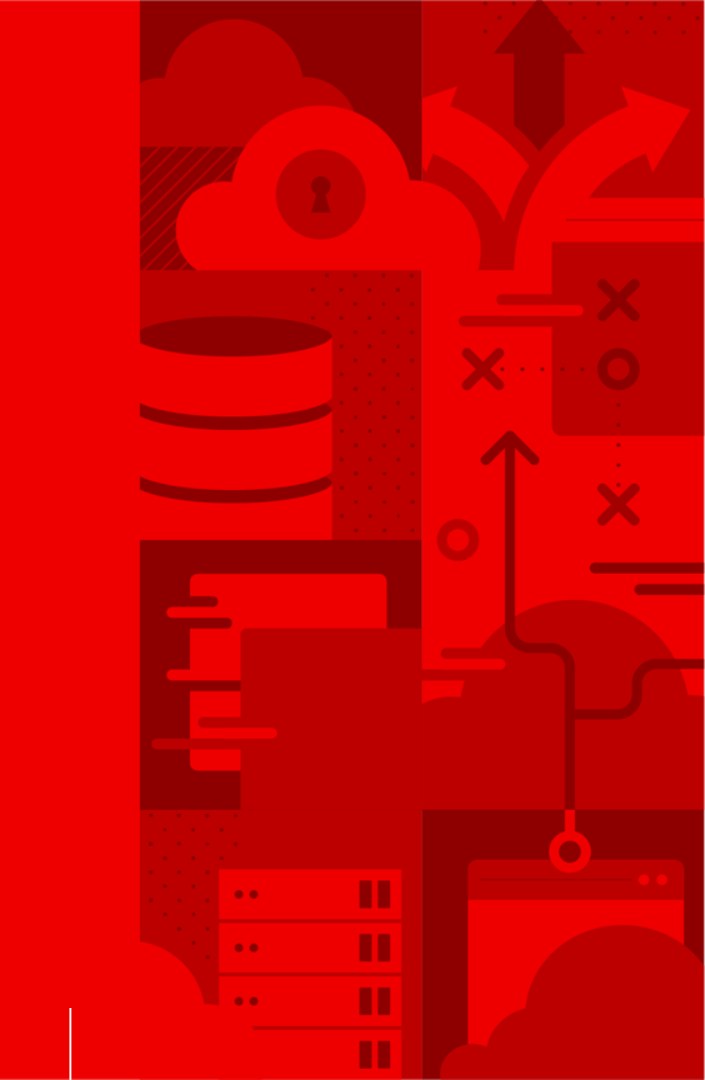
A man with short dark hair and glasses, wearing a black long-sleeved shirt, is sitting on a grey couch. He is holding a white smartphone in his hands. A woman with long brown hair, wearing a black shirt with white polka dots and white pants, is sitting next to him, looking down at the phone. They are both focused on the device. The background is a bright, out-of-focus window showing a cityscape.

Omnichannel implies integration and orchestration of channels such that the experience is of engaging across all the channels someone chooses to use.



Tackling the challenges

- Data management, security, and user access
- Multiple protocol and language support through different integration technologies
- Distributed deployments, non-centralized integration
- Training developers to deliver in a digital integration architecture
- Training operations to manage and monitor a digital integration landscape

A vertical red graphic on the left side of the slide. It contains various white icons representing technology: a cloud with a keyhole, a database cylinder, a server rack, a monitor, and several arrows pointing in different directions. There are also 'X' and 'O' symbols and a circuit-like line.

Portfolio Architecture

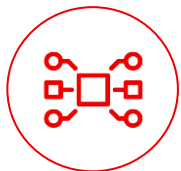
Generic view of the solution architecture

Omnichannel Customer Experience Introduction

Business Drivers



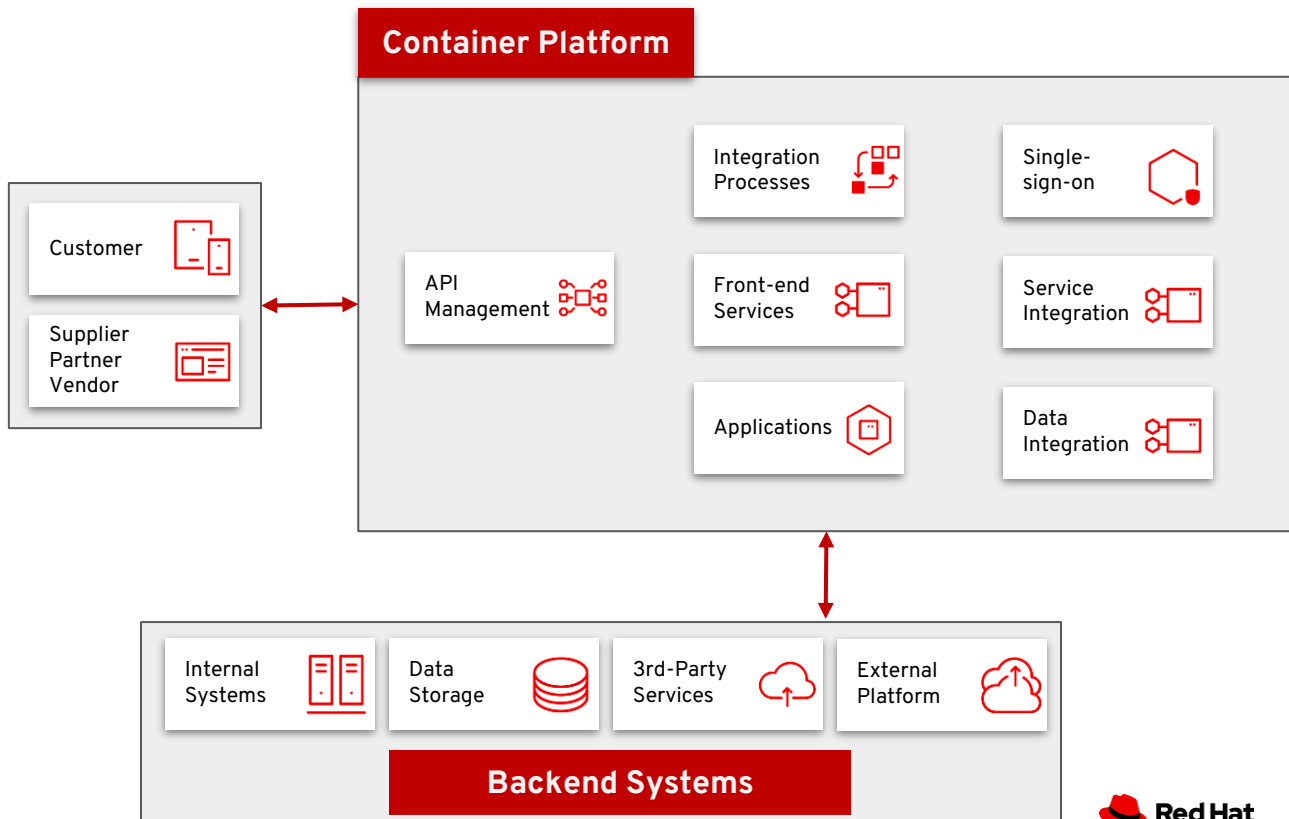
Data management, security, and user access



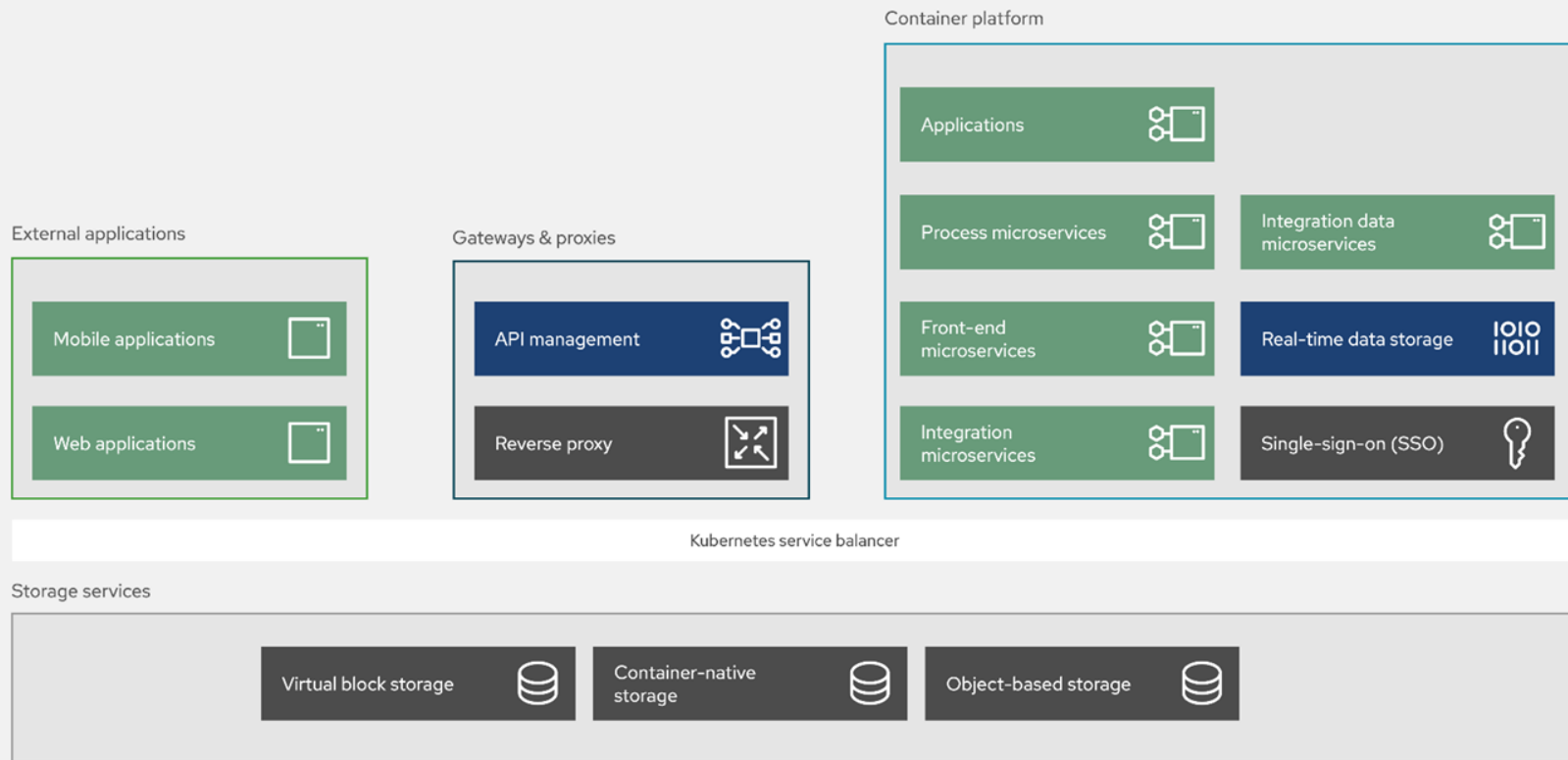
Multiple protocol support through different integration technologies



Distributed deployments, non-centralized integration

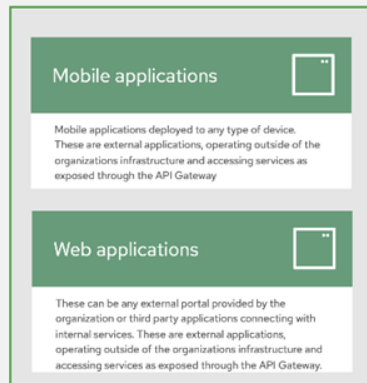


Logical diagram - Omnichannel customer experience

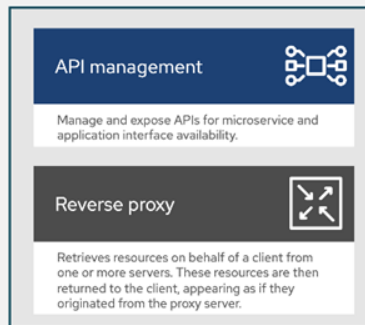


Logical diagram - Omnichannel customer experience (details)

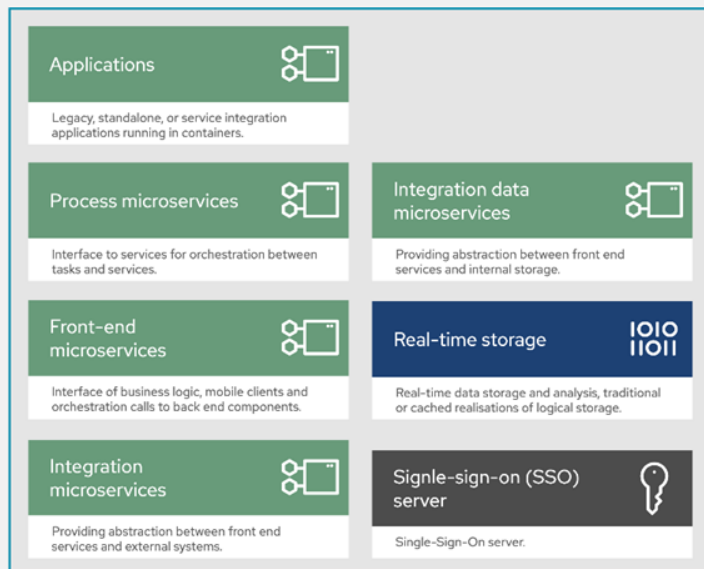
External applications



Gateways & proxies

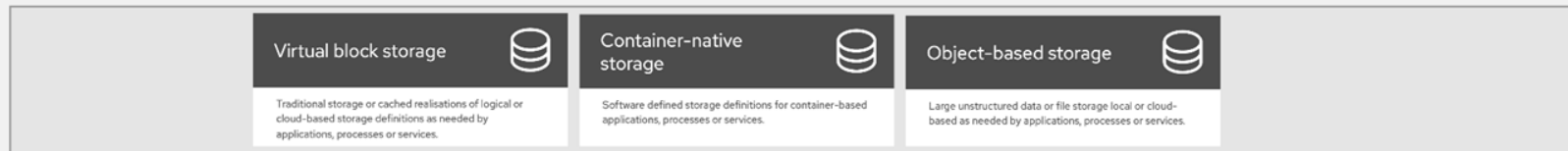


Container platform

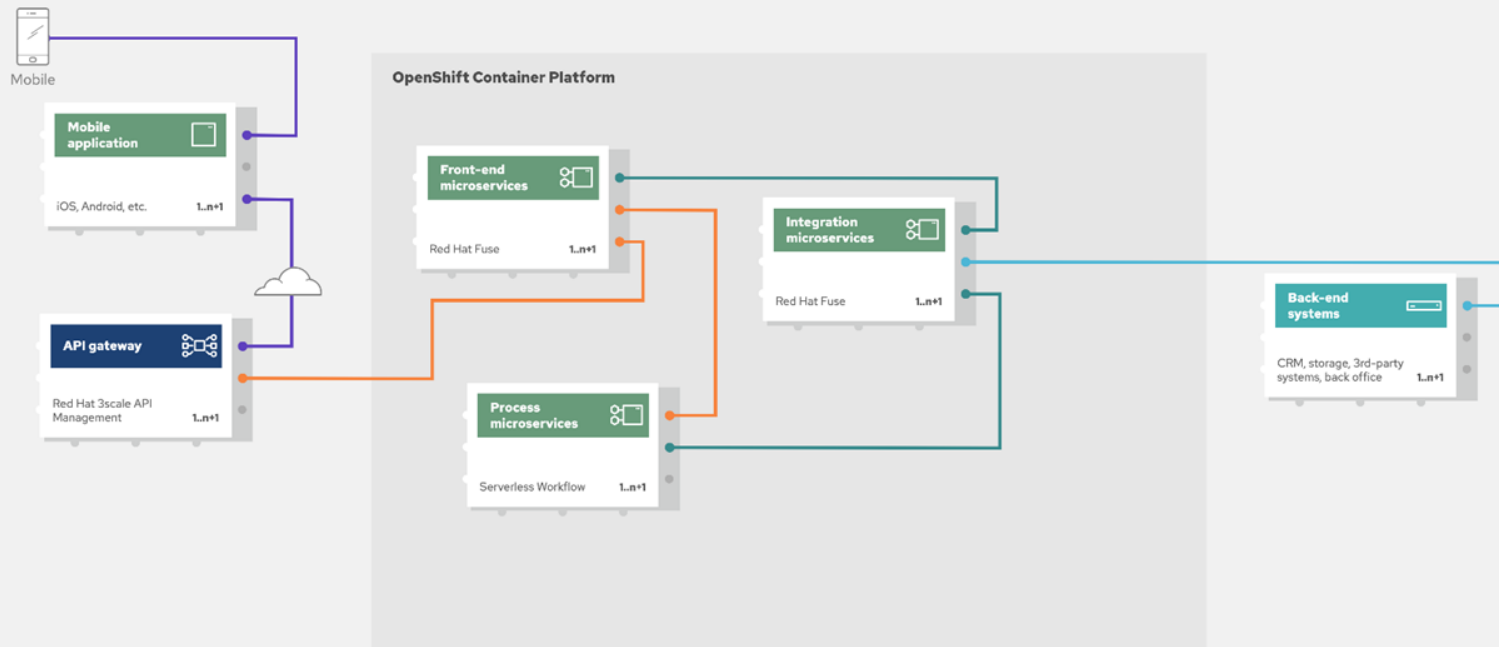


Kubernetes service balancer

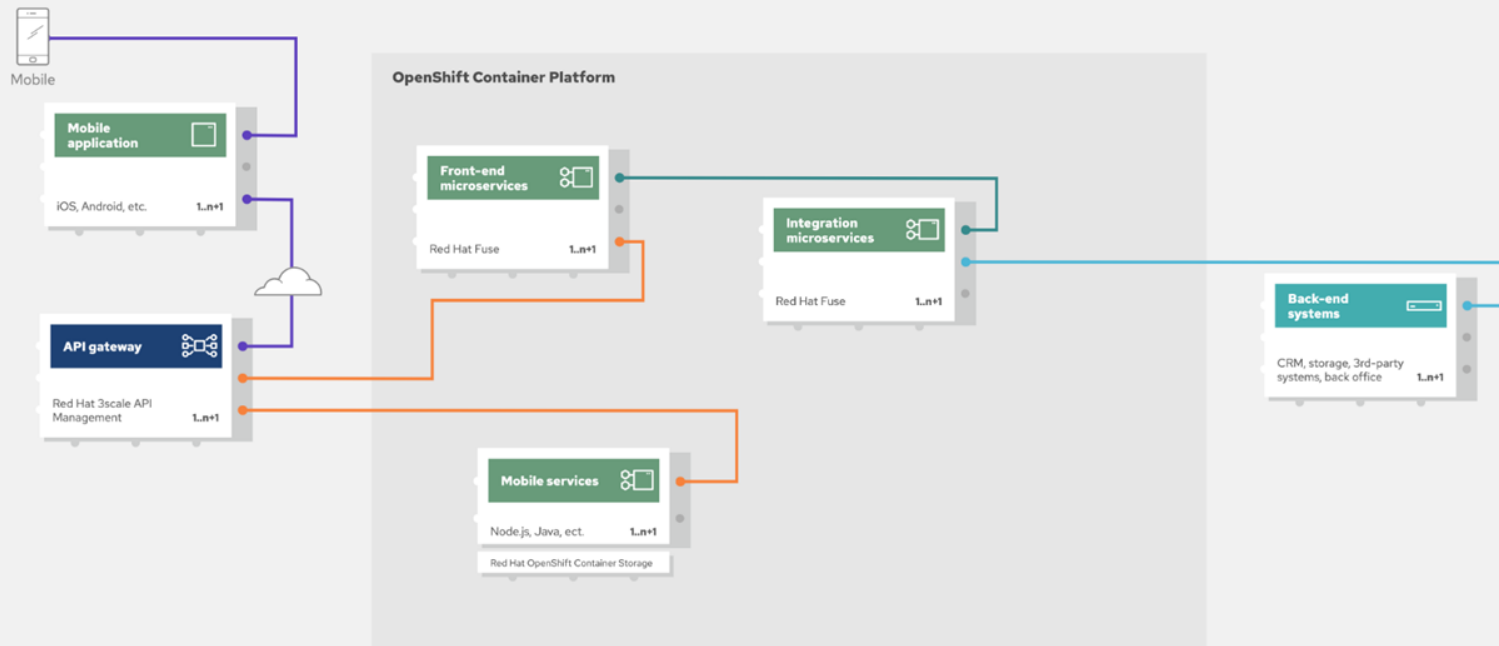
Storage services



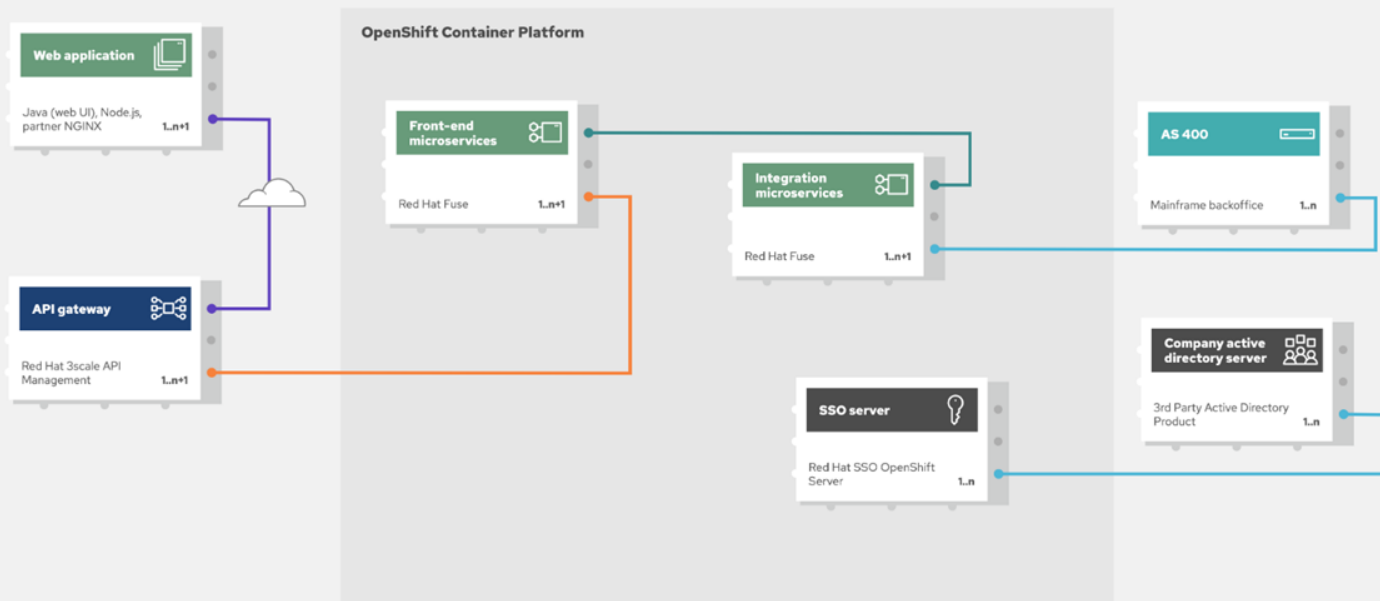
Schematic diagram - Process integration example



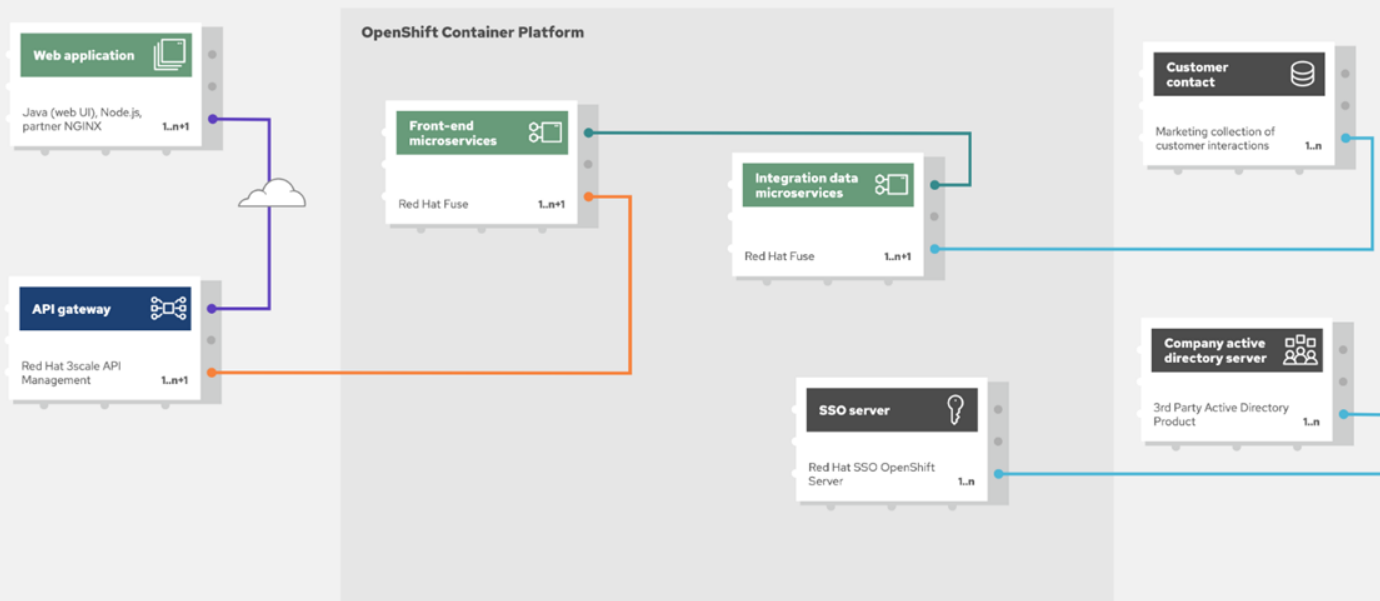
Schematic diagram - Mobile integration example



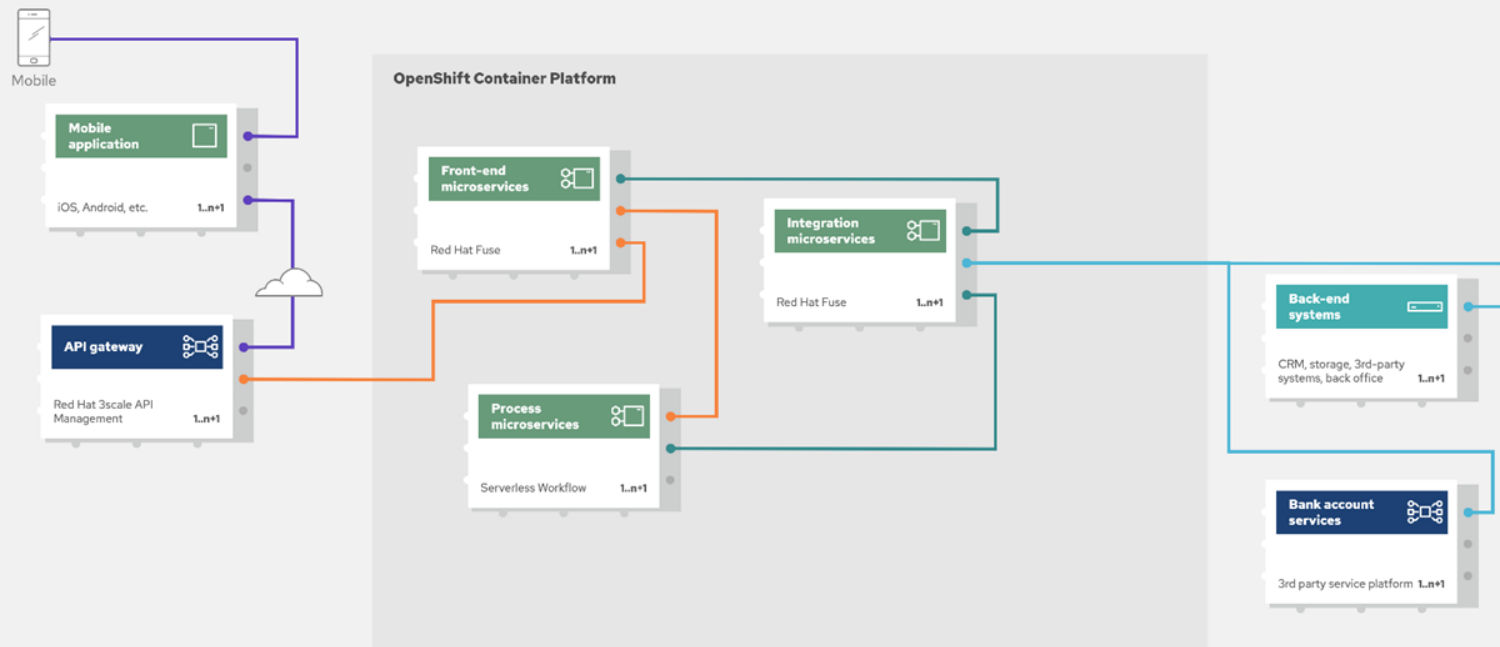
Schematic diagram - Integration service example



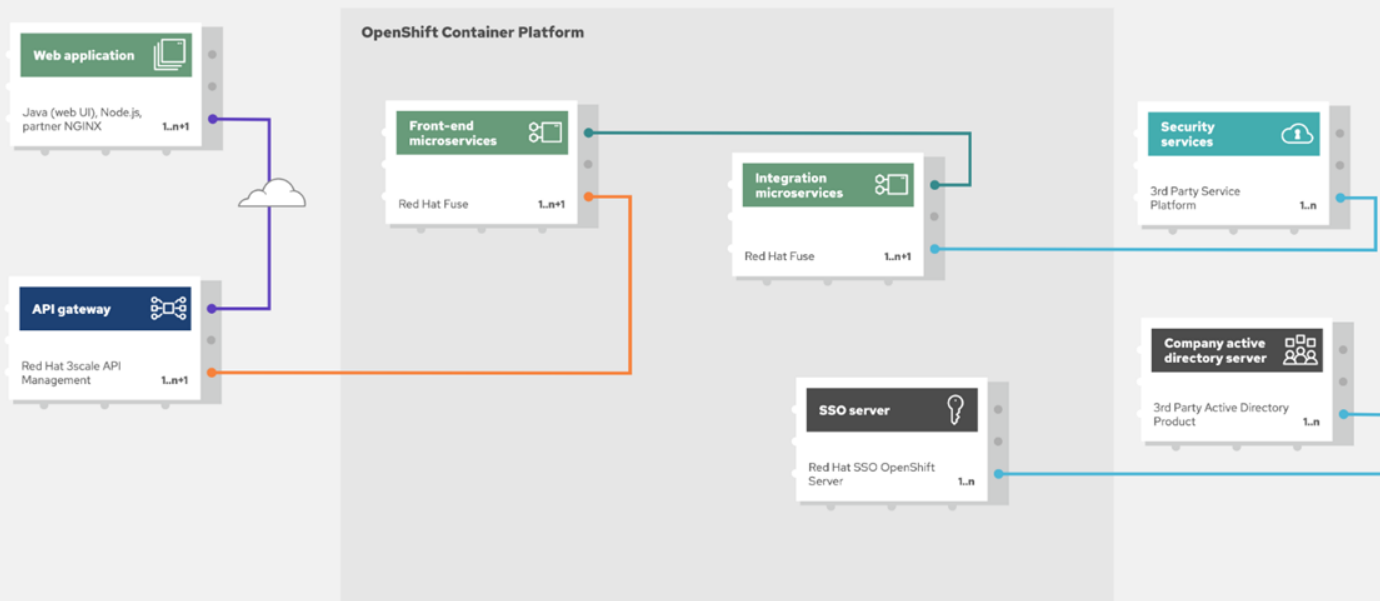
Schematic diagram - Integration data service example

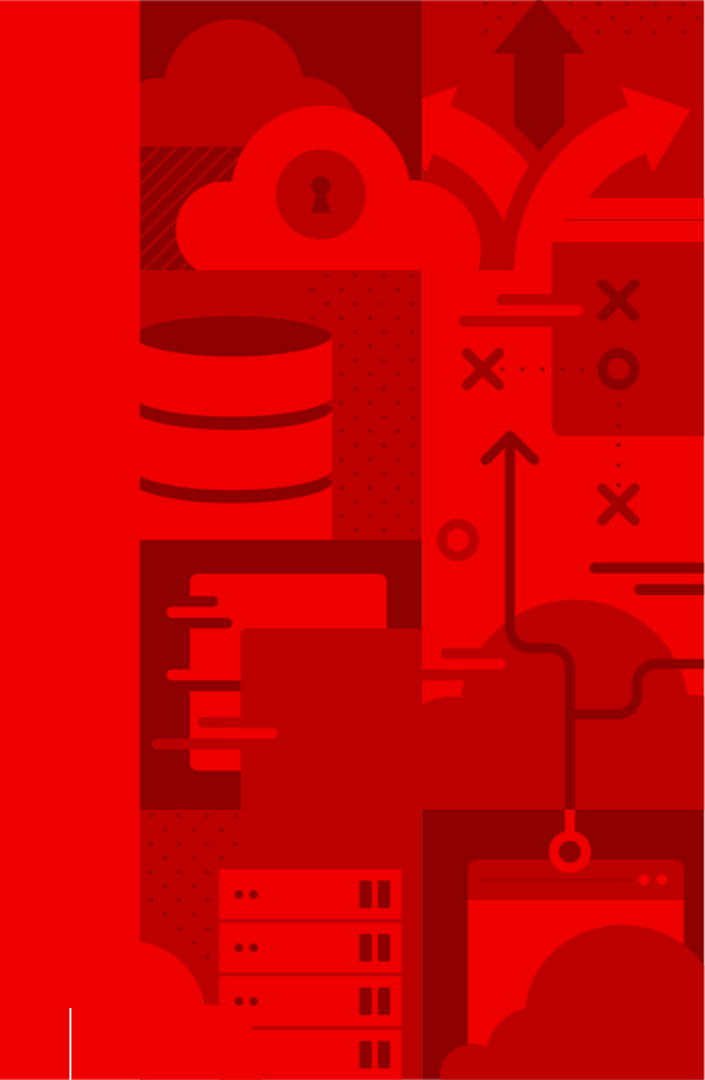


Schematic diagram - Process integration with 3rd party services example



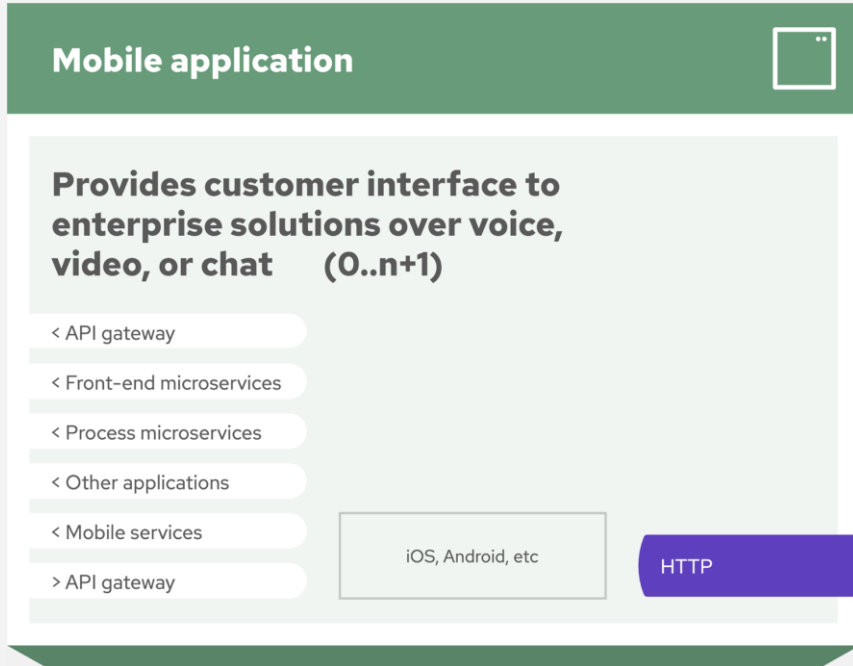
Schematic diagram - Integration 3rd party service example





APPENDIX WITH ELEMENT DETAILS

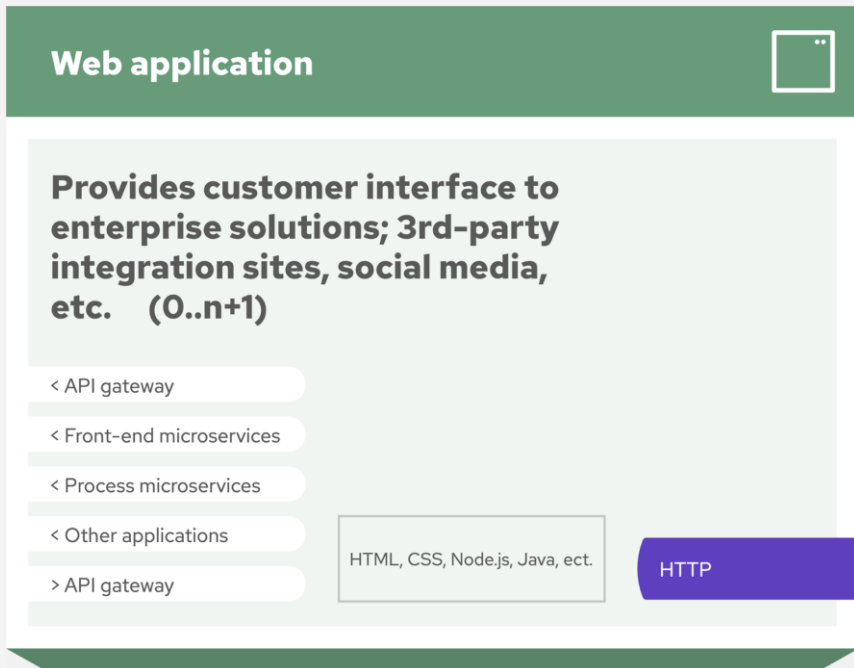
DETAIL: MOBILE APPLICATION



Application for mobile devices:

- Provides customer interface to enterprise solutions
- Voice, video or chat features
- Contains service calls to:
 - API Gateway for:
 - Frontend Microservices
 - Process Microservices
 - Other Applications (aggregated microservices)
 - Eventual mobile specific services (push notifications, usage analytics, etc)

DETAIL: WEB APPLICATION



Web sites from enterprise or third-party sites using provided services:

- Provides customer interface to enterprise solutions
- Third party integration sites, retail sites, social media, and more
- Contains service calls to:
 - API Gateway for:
 - Frontend Microservices
 - Process Facade Microservices
 - Other Applications (aggregated microservices)

DETAIL: API MANAGEMENT

API management



Provides service interfaces to applications and other microservices; provides scalability, reliability and interface usage metrics (1..n+1)

> Front-end microservices

> Process microservices

> Other applications

< External requests

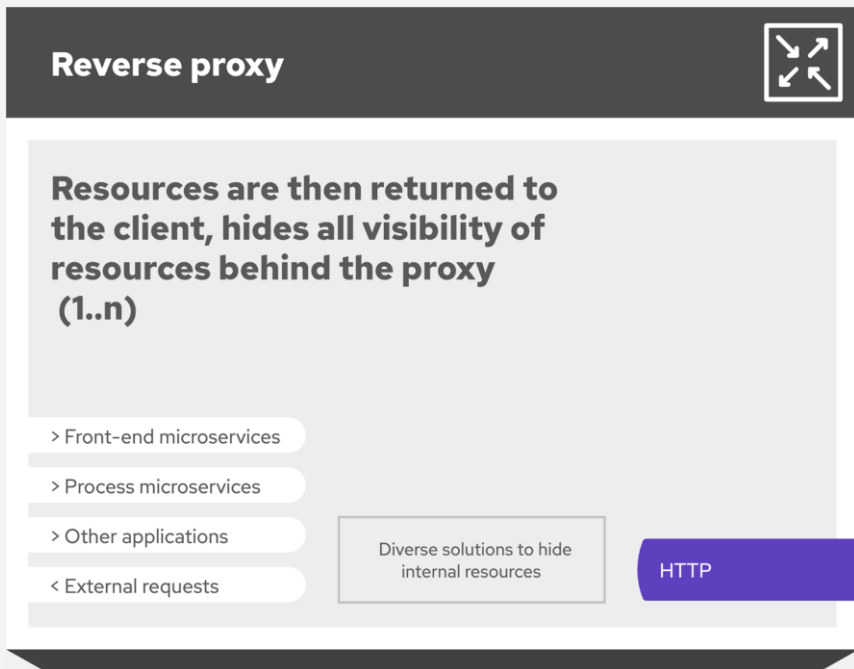
API management platform

HTTP

Platform to manage and expose APIs for microservices

- Provides service interfaces to applications and other microservices
- Provides scalability, reliability and interface usage metrics
- Exposes interfaces from:
 - Frontend Microservices
 - Process Microservices
 - Other Applications (aggregated microservices)

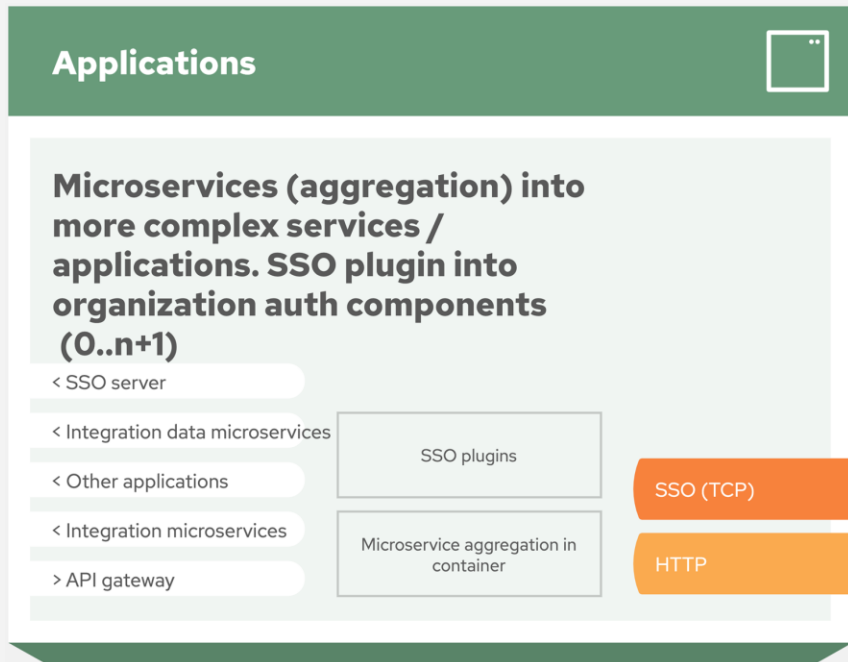
DETAIL: REVERSE PROXY



Retrieves resources on behalf of a client

- Resources are then returned to the client
- Hides all visibility of resources behind the proxy
- Manages external traffic to:
 - Frontend Microservices
 - Process Microservices
 - Other Applications (aggregated microservices)

DETAIL: APPLICATIONS



Legacy, standalone, or service integration applications running in containers.

- Microservices (aggregation) into more complex services / applications
- SSO plugin into organization auth components
- Receives calls from:
 - API Gateway
- Potential services calls to:
 - Integration (Data) Microservices
 - Applications
 - SSO Server

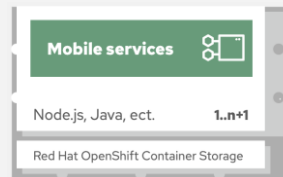
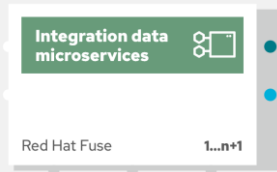
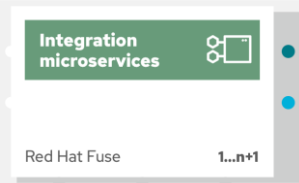


APPLICATION – CUSTOMER CONTACT



Customer Contact Service

Called every time customer touched



This service details a customer interaction and updates the call center application's view of the customer.



1... n+1

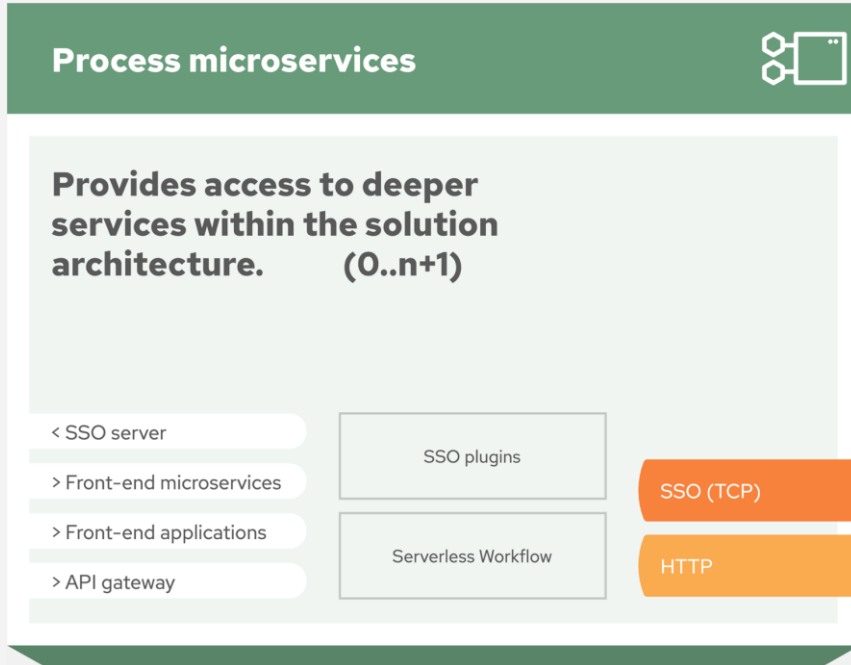


CNS

SSO



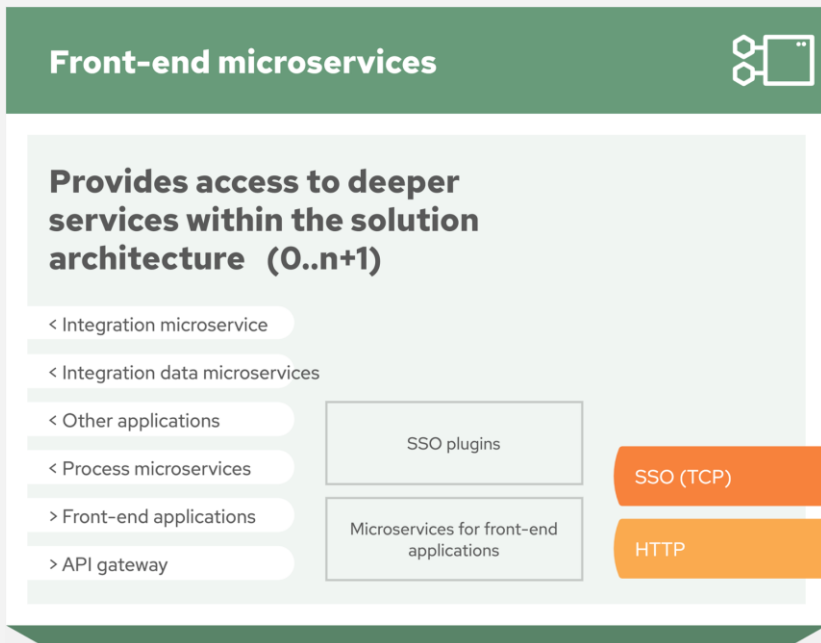
DETAIL: PROCESS MICROSERVICES



Services used to access process automation (processes):

- Provides access to deeper services within the solution architecture
- SSO plugin into organization auth components
- Receives calls from:
 - Frontend applications
 - Applications (microservices)
 - Frontend microservices
- Potential services calls to:
 - SSO server

DETAIL: FRONT-END MICROSERVICE



Services used in front-end apps looking to work with microservice infrastructure provided:

- Provides access to deeper services within the solution architecture
- SSO plugin into organization auth components
- Receives calls from:
 - Front-end applications
- Potential services calls to:
 - Integration (Data) Microservices
 - Process Microservices
 - SSO Server
 - Applications

DETAIL: INTEGRATION MICROSERVICES

Integration microservices



Provides access to components within the solution architecture. (0..n+1)

< Integration data microservices

< SSO server

> Front-end microservices

< Process microservices

> Other applications

SSO server

Integrate backend systems and
3rd-party services

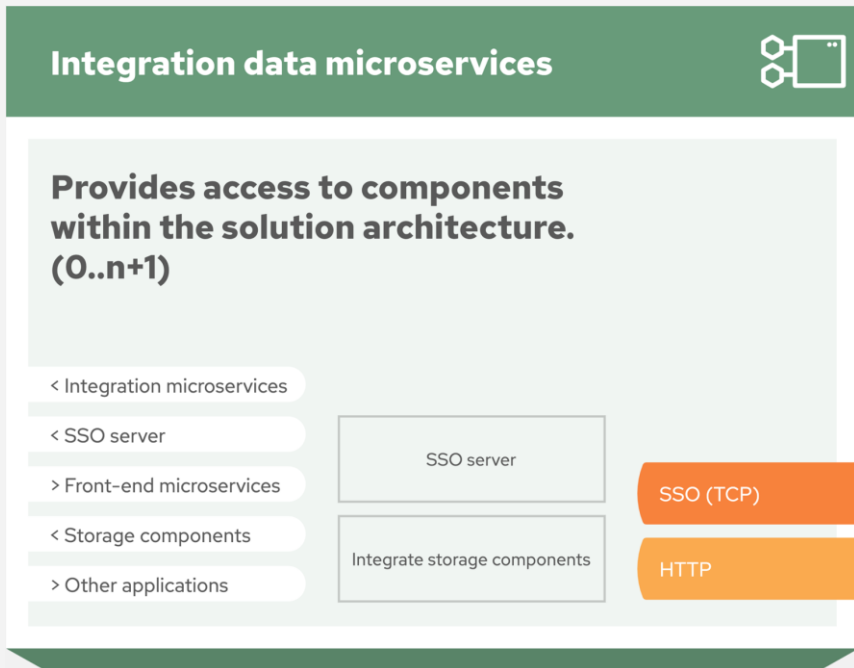
SSO (TCP)

HTTP

Services integrating all manner of backend systems and 3rd-party services:

- Provides access to components within the solution architecture
- SSO plugin into organization auth components
- Receives calls from:
 - Frontend Microservices
 - Process Server
 - Applications
- Potential services calls to:
 - Integration Data Microservices
 - Process Microservices
 - SSO Server

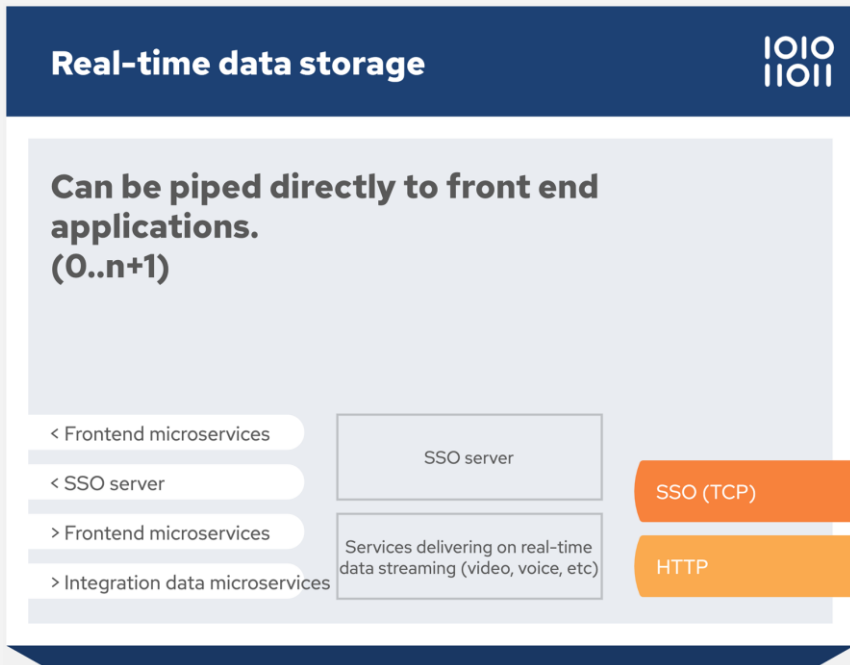
DETAIL: INTEGRATION DATA MICROSERVICES



Services integrating storage components:

- Provides access to components within the storage solution architecture
- SSO plugin into organization auth components
- Receives calls from:
 - Frontend Microservices
 - Applications
 - Integration Microservices
- Potential services calls to:
 - Storage components
 - SSO Server

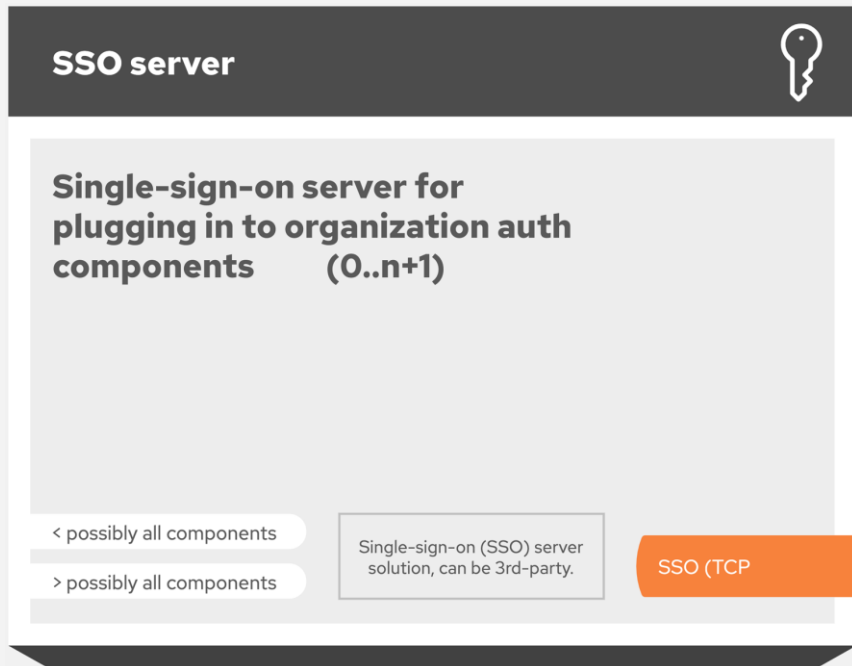
DETAIL: REAL-TIME DATA STORAGE



Services delivering on real-time data streaming (video, voice, ect).

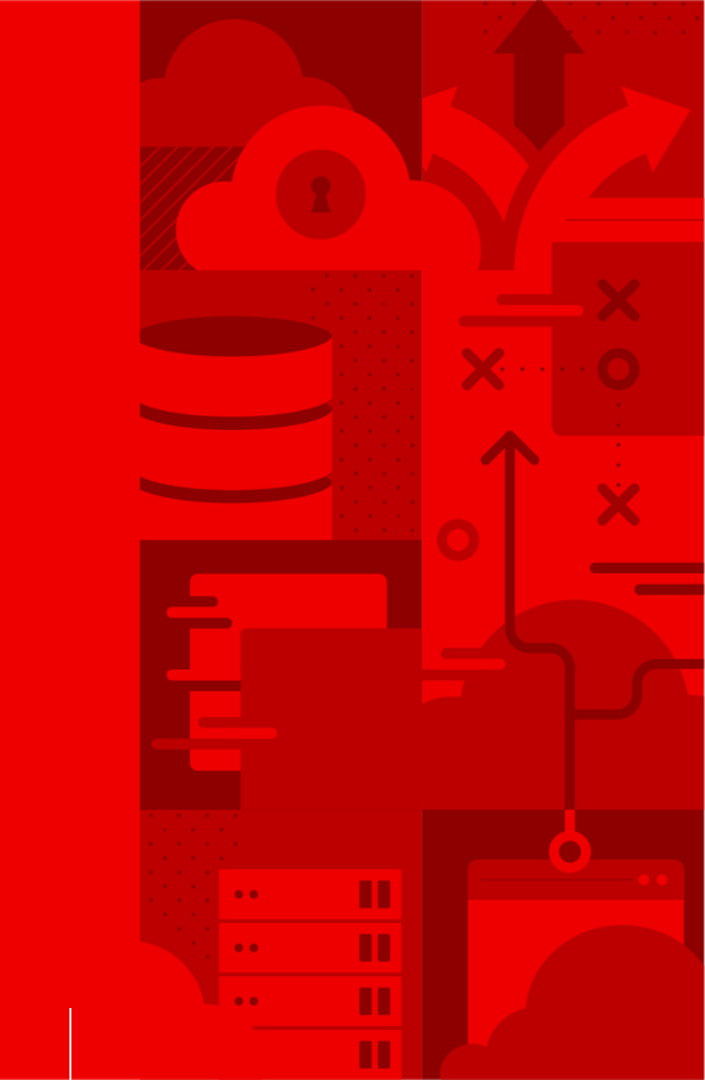
- Can be pipelined directly to front end applications.
- SSO plugin into organization auth components
- Receives calls from:
 - Integration Data Microservices
 - Frontend Microservices
- Potential services calls to:
 - Frontend Microservices
 - SSO Server

DETAIL: SSO SERVER



Single-sign-on (SSO) server solution, can be 3rd-party.

- Single Sign On server for plugging in to organization auth components
- Receives calls from:
 - Possibly all architecture components
- Potential services calls to:
 - Possibly all architecture components



Thank You

End of the public story